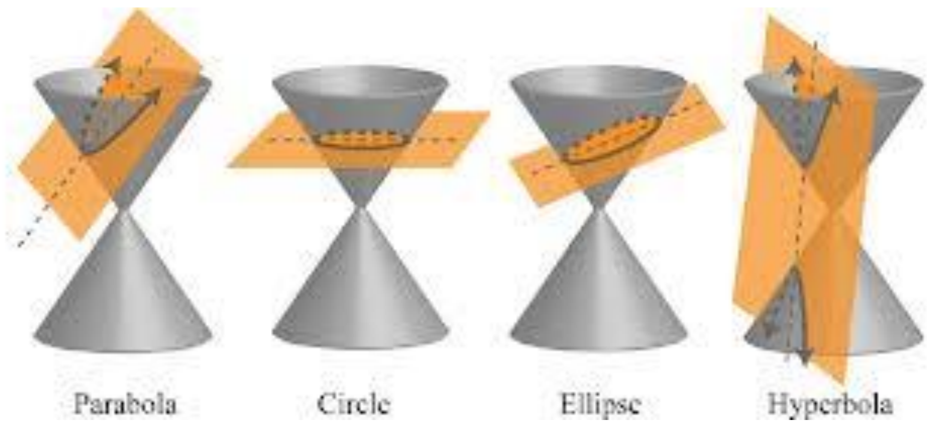
5.4 Summary of Conic Sections

If a plane passes through a double right circular cone there are four possible resulting curves. These curves are the **parabola**, **circle**, **ellipse**, and **hyperbola**. These curves are called **conic sections** and are shown below.



The Equation of a Parabola: (opening up or down)
The parabola

$$y = a(x-h)^2 + k$$

has focus $(h, k+p)$

and directrix $y = k - p$ where $a = \dfrac{1}{4p}$ and (h,k) is the vertex.

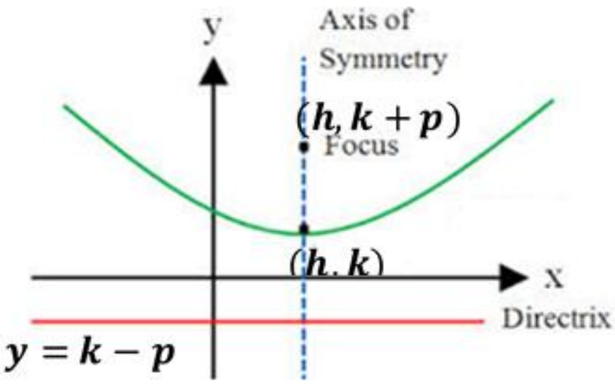
The Equation of a Parabola: (opening left or right)
The parabola

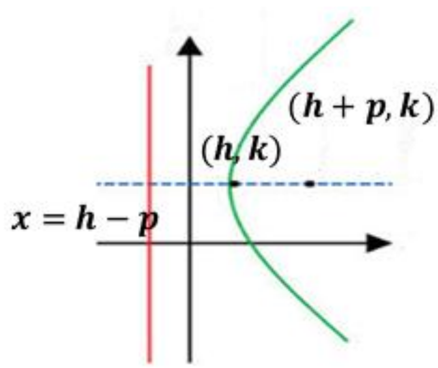
$$x = a(y-k)^2 + h$$

has focus $(h+p, k)$

and directrix $x = h - p$ where $a = \dfrac{1}{4p}$ and (h,k) is the vertex.

Note: If $a > 0$ then the parabola opens to the right. If $a < 0$ then the parabola opens to the left..





<u>The Equation of an Ellipse: (with a *horizontal* major axis and center at the origin)</u>

The ellipse with center at the origin and equation

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, \qquad a>b$$

has vertices $(\pm a,\, 0)$, endpoints of the minor axis $(0,\, \pm b)$, and foci $(\pm c,\, 0)$.

where $c^2=a^2-b^2$

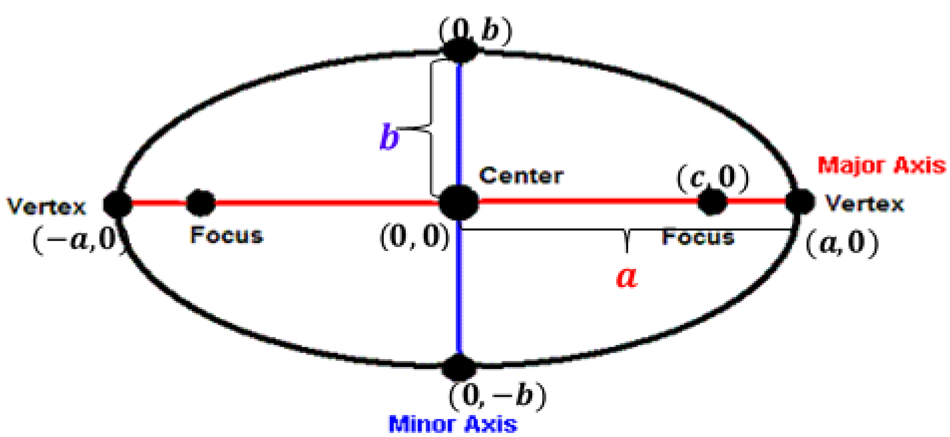
<u>The Equation of an Ellipse: (with a *vertical* major axis and center at the origin)</u>

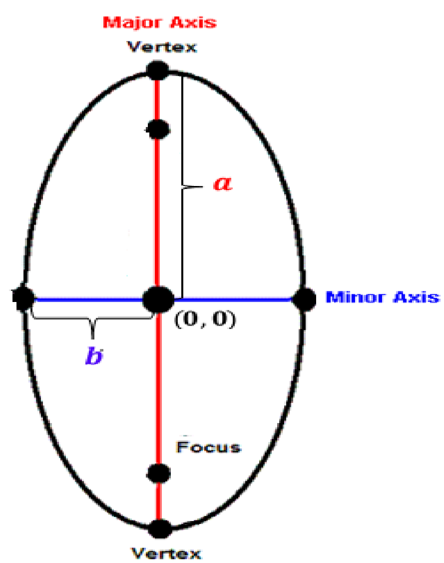
The ellipse with center at the origin and equation

$$\frac{x^2}{b^2}+\frac{y^2}{a^2}=1, \qquad a>b$$

has vertices $(0, \pm a)$, endpoints of the minor axis $(\pm b, 0)$, and foci $(0, \pm c)$.

where $c^2=a^2-b^2$





<u>Equation for a Circle</u>

The standard equation of a circle with center (h,k) and radius r $(r>0)$ is

$$(x-h)^2+(y-k)^2=r^2$$

Note: By dividing both sides of this equation by r^2 we obtain $\frac{(x-h)^2}{r^2}+\frac{(y-k)^2}{r^2}=1$. So clearly a circle is an ellipse with major and minor axes of the same length.

The Equation of a Hyperbola: (opening left or right)

The equation of a hyperbola with center at the origin with foci $(\pm c, 0)$ and x-intercepts $(\pm a, 0)$ is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

where $c^2 = a^2 + b^2$

The differences of the distances from any point of the hyperbola to the two foci is *2a*.

The Equation of a Hyperbola: (opening up or down)

The equation of a hyperbola with center at the origin with foci $(0, \pm c)$ and x-intercepts $(0, \pm a)$ is

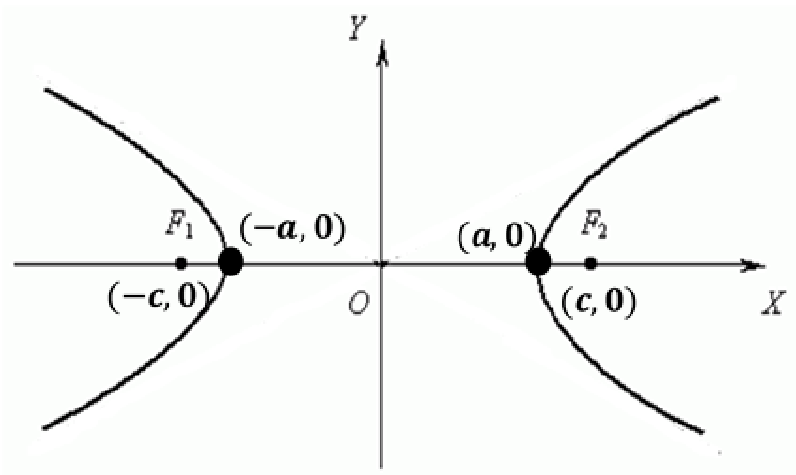
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

where $c^2 = a^2 + b^2$

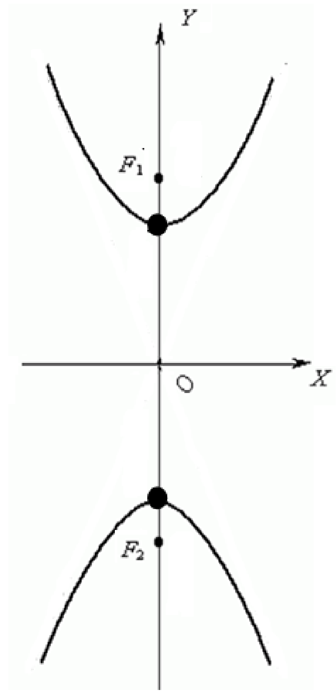
The differences of the distances from any point of the hyperbola to the two foci is *2a*.

Note: With hyperbolas it is not important whether a or b is the larger value. We always think of a as the value beneath y and b as the value beneath x.

Note: If $a > 0$ *then the parabola opens up. If* $a < 0$ *then the parabola opens down.*



Example 1: Determine what type of conic section represented by the equation $9x^2 - 4y^2 - 72x + 8y + 176 = 0$ and sketch a rough graph of the conic.



Example 2: Determine what type of conic section represented by the equation $9x^2 - 54x - 19 = 100 - 50y - 25y^2$ and sketch a rough graph of the conic.

Example 3: Determine what type of conic section represented by the equation $x^2 + 4x + y^2 - 2y = 11$ and sketch a rough graph of the conic.

Example 4: Determine what type of conic section represented by the equation $2y + x - y^2 = 0$ and sketch a rough graph of the conic.